PageRank

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1 PageRank Algorithm

The PageRank algorithm is a well-known method used to rank web pages based on their importance, and it was originally developed to improve search engine results by looking at the link structure of the web instead of just focusing on keywords.

In this approach, we can model the web as a directed graph where each node represents a web page and every directed edge represents a hyperlink from one page to another. The core idea is that importance is shared: if a page is linked by many other "important" pages, then that page should also be considered important itself.

1.1 Matrix formulation

Let N be the total number of web pages. The link structure of the web is modeled by a directed graph and represented by the adjacency matrix $B \in R^{N \times N}$, defined as

$$B_{ij} = \begin{cases} 1 & \text{if page } j \text{ contains a link to page } i, \\ 0 & \text{otherwise.} \end{cases}$$

For each page j, let n_j denote the number of outgoing links. If $n_j > 0$, the adjacency matrix is normalized column-wise to obtain the matrix A, whose entries are

$$A_{ij} = \frac{B_{ij}}{n_j}.$$

The matrix A is column-stochastic, meaning that the entries of each column sum to one.

1.2 Dangling nodes

Some pages may have no outgoing links. These pages are called dangling nodes and produce columns of zeros in the matrix A. To avoid this problem, the probability associated with dangling nodes is redistributed uniformly among all pages.

1.3 PageRank definition

The PageRank vector x assigns an importance score to each page and is defined as the solution of

$$x = dAx + d\left(\sum_{j \in D} x_j\right) s + ms,$$

where d is the damping factor, $m = 1 - d$, D denotes the set of dangling nodes, and $s = \frac{1}{N}\mathbf{1}$ is the uniform probability vector.

The vector x is normalized so that the sum of its components is equal to one, allowing it to be interpreted as a probability distribution over the set of pages.

1.4 Power method

Starting from a uniform initial vector, the PageRank vector is updated iteratively by combining three contributions: the link structure of the web, the redistribution of probability from dangling nodes, and the teleportation term. At each iteration, a new approximation of the PageRank vector is computed until convergence. The stopping criterion is based on the relative difference between two consecutive iterates. This iterative scheme is easy to implement and is well suited for large sparse matrices.

2 Implementation

The PageRank algorithm described in the previous section has been implemented in MATLAB. The web graph is loaded from a data file and represented using sparse matrices in order to reduce memory usage. First, the adjacency matrix B is constructed from the dataset, following the convention $B_{ij} = 1$ if page j links to page i. Then, the column-stochastic matrix A is obtained by normalizing each column with respect to the number of outgoing links. Dangling nodes are detected and handled by redistributing their probability mass uniformly over all pages, consistently with the PageRank definition. The PageRank vector is computed using the power method, starting from a uniform initial vector. At each iteration, the contribution of link-following, dangling nodes, and teleportation is added. The iteration stops when the relative error between two consecutive iterates is below a fixed tolerance or when a maximum number of iterations is reached. Finally, the resulting PageRank vector is normalized and the pages with the highest scores are printed.

3 Results

3.1 Dataset results

The PageRank algorithm has been applied to a real web dataset composed of $N = 6012$ nodes, each representing a web page, and a set of directed edges representing hyperlinks between pages. The damping factor has been set to $d = 0.85$, which is the standard value commonly used.
The power method was initialized with a uniform probability vector and iterated until convergence. The stopping criterion was based on the relative error between two consecutive iterates, with tolerance $\varepsilon = 10^{-6}$.

The power method converged in a finite number of iterations. This confirms the effectiveness of the PageRank formulation and the numerical stability of the algorithm when applied to large sparse graphs.

Rank	Node	PageRank score
1	2	0.01987875
2	37	0.00928762
3	38	0.00861039
4	61	0.00806503
5	52	0.00802656
6	43	0.00716464
7	425	0.00658278
8	27	0.00598921
9	28	0.00557174
10	4023	0.00445247
11	29	0.00438508
12	5254	0.00377793
13	3227	0.00374159
14	40	0.00352556
15	3834	0.00339769
16	822	0.00333746
17	4075	0.00333375
18	73	0.00331817
19	132	0.00322873
20	81	0.00309215

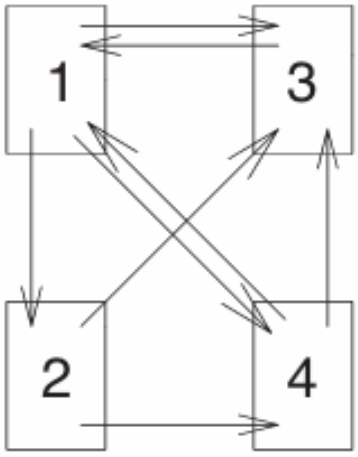
Table 1: Top 20 pages ranked by PageRank score for the dataset with 6012 nodes.

3.2 Graph example A

The PageRank algorithm has been applied to a small directed graph with four nodes. Table 2 reports the PageRank values obtained for this example, while Figure 3.2 shows the corresponding directed graph.

Rank	Node	PageRank score
1	1	0.36815065
2	3	0.28796160
3	4	0.20207834
4	2	0.14180941

Table 2: PageRank values



figureGraph A

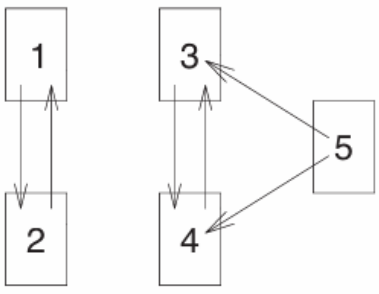
The PageRank values in Table 2 reflect the structure of the directed graph shown in Figure. Node 1 attains the highest score since it receives several incoming links, including links from nodes involved in cycles, which reinforce its importance. Node 3 ranks second because it is linked by relevant nodes, while nodes 4 and 2 obtain lower PageRank values due to their weaker incoming connectivity. Overall, the ranking highlights the ability of PageRank to capture the structural importance of nodes in a directed graph.

3.3 Graph example B

The PageRank algorithm has been applied to a small directed graph with five nodes. Table 3 reports the PageRank values obtained for this example, while Figure 3.3 shows the corresponding directed graph.

Rank	Node	PageRank score
1	3	0.28500000
2	4	0.28500000
3	1	0.20000000
4	2	0.20000000
5	5	0.03000000

Table 3: PageRank values.



Graph B

The PageRank values in Table 3 can be explained by the structure of the directed graph and

by its connected components. The graph contains two strongly connected pairs of nodes, $\{1, 2\}$ and $\{3, 4\}$, and a node 5 that has only outgoing links. Nodes 3 and 4 have the highest and equal PageRank values because they are part of a strongly connected component and also receive probability from node 5. Nodes 1 and 2 form another strongly connected component, but since they are not connected to the rest of the graph, they obtain lower PageRank scores. Node 5 has the lowest PageRank value because it does not receive any incoming links and only passes its probability to other nodes. This example shows that PageRank gives higher scores to nodes that belong to well-connected parts of the graph and lower scores to nodes with no incoming links.

4 Exercise solutions

4.1 Exercise 8

A matrix $A \in R^{n \times n}$ is called *column-stochastic* if

$$a_{ij} \geq 0 \quad \text{for all } i, j, \qquad \text{and} \qquad \sum_{i=1}^{n} a_{ij} = 1 \quad \text{for every } j = 1, \ldots, n. \tag{1}$$

Let $A, B \in R^{n \times n}$ be column-stochastic matrices. Then the product $C = AB$ is column-stochastic. We show that C satisfies both defining properties of a column-stochastic matrix.

Non-negativity: For any $i, j \in \{1, \ldots, n\}$, the (i, j)-entry of C is given by

$$c_{ij} = (AB)_{ij} = \sum_{k=1}^{n} a_{ik}\, b_{kj}. \tag{2}$$

Since A and B are column-stochastic, all their entries are non-negative. Therefore, each product $a_{ik}b_{kj}$ is non-negative, and hence

$$c_{ij} \geq 0 \quad \text{for all } i, j.$$

Column sums: Fix an index $j \in \{1, \ldots, n\}$ and consider the j-th column of C. Summing (2) over i, we obtain

$$\begin{aligned} \sum_{i=1}^{n} c_{ij} &= \sum_{i=1}^{n} \sum_{k=1}^{n} a_{ik}\, b_{kj} \\ &= \sum_{k=1}^{n} b_{kj} \left(\sum_{i=1}^{n} a_{ik} \right), \end{aligned} \tag{3}$$

where the order of summation has been exchanged, which is allowed since the sums are finite. The inner sum $\sum_{i=1}^{n} a_{ik}$ represents the sum of the entries of the k-th column of A and, since A is column-stochastic, it is equal to 1 for every k. Therefore,

$$\sum_{i=1}^{n} c_{ij} = \sum_{k=1}^{n} b_{kj}.$$

Since B is column-stochastic, the sum of the entries of its j-th column is equal to 1, and thus

$$\sum_{i=1}^{n} c_{ij} = 1 \qquad \text{for all } j = 1, \ldots, n.$$

Combining this result with the non-negativity of the entries of C, we conclude that C is column-stochastic.

4.2 Exercise 10

4.2.1 Show that $(A^2)_{i,j} > 0$ if and only if page i can be reached from page j in exactly two steps.

In a directed graph representing web pages: $A_{ij} > 0$ means there is a direct link (1 step) from page j to page i. $A_{ij} = 0$ means there is no direct link from j to i.

By the definition of matrix multiplication, the entry in the i-th row and j-th column of A^2 is:

$$(A^2)_{ij} = \sum_{k=1}^{n} A_{ik} A_{kj}$$

Forward Direction: If $(A^2)_{ij} > 0$, then there is a 2-step path
If $(A^2)_{ij} > 0$, there must exist at least one index k such that:

$$A_{ik} A_{kj} > 0$$

For this product to be positive, both factors must be positive, so there is a link from page j to page k (Step 1) and a link from page k to page i (Step 2). The existence of such a k proves there is a path $j \to k \to i$, which is exactly two steps.

Backward Direction: If there is a 2-step path, then $(A^2)_{ij} > 0$
If page i can be reached from page j in exactly two steps, there must be some intermediate page k such that: There is a link from j to k ($\implies A_{kj} > 0$). There is a link from k to i ($\implies A_{ik} > 0$). Since both A_{ik} and A_{kj} are positive, their product $A_{ik}A_{kj}$ must be positive. In the summation $(A^2)_{ij} = \sum_{m=1}^{n} A_{im}A_{mj}$, one of the terms in the sum (where $m = k$) is positive. Since all other terms are ≥ 0, the entire sum must be greater than zero:

$$(A^2)_{ij} \geq A_{ik} A_{kj} > 0$$

This logic can be extended via induction to show that $(A^n)_{ij} > 0$ if and only if there is a path of length n from j to i.

4.2.2 Show more generally that $(A^p)_{ij} > 0$ if and only if page i can be reached from page j in EXACTLY p steps.

Assume the statement is true for some integer $p = k$. That is: $(A^k)_{ij} > 0 \iff$ there is a path of length k from j to i. Now, we must show it is true for $p = k + 1$. We look at the entry $(A^{k+1})_{ij}$ using the definition of matrix multiplication:

$$(A^{k+1})_{ij} = (A^k \cdot A)_{ij} = \sum_{m=1}^{n} (A^k)_{im} A_{mj}$$

Forward Direction: If $(A^{k+1})_{ij} > 0$, then a path of length $k+1$ exists.
Since all entries in A and A^k are non-negative, for the sum to be positive, there must exist at least one index m such that:

$$(A^k)_{im} A_{mj} > 0$$

This implies two things must be true simultaneously:

- $A_{mj} > 0$: There is a 1-step path from j to m
- $(A^k)_{im} > 0$: By our inductive hypothesis, there is a path of exactly k steps from m to i. these, we have a path: $j \xrightarrow{\text{1 step}} m \xrightarrow{\text{k steps}} i$. The total number of steps is $1+k$, confirming a path of exactly $k+1$ steps.

Backward Direction: If a path of length $k+1$ exists, then $(A^{k+1})_{ij} > 0$.
If there is a path of length $k+1$ from j to i, there must be some intermediate node m which is the first stop after j, i.e. here is a link from j to m ($\implies A_{mj} > 0$). There is a remaining path of length k from m to i ($\implies (A^k)_{im} > 0$ by the hypothesis). Since both terms are positive, their product $(A^k)_{im} A_{mj}$ is positive. Because this product is one of the terms in the summation for $(A^{k+1})_{ij}$, the total sum must be greater than zero.

4.2.3 Argue that

$$(I + A + A^2 + ... + A^p)_{ij} > 0$$

if and only if page i can be reached from page j in p or fewer steps (note p = 0 is a legitimate choice—any page can be reached from itself in zero steps!)

To understand why the sum of these matrices represents reachability, we can look at the sum entry-by-entry.
Let S_p be the matrix defined by:

$$S_p = I + A + A^2 + \cdots + A^p$$

The entry $(S_p)_{ij}$ is the sum of the corresponding entries from each individual matrix:

$$(S_p)_{ij} = (I)_{ij} + (A)_{ij} + (A^2)_{ij} + \cdots + (A^p)_{ij}$$

As established previously, all entries of the adjacency matrix A are non-negative ($A_{ij} \geq 0$). Consequently, all powers of A ($A^2, A^3, \dots$) also contain only non-negative entries.In a sum of non-negative terms, the total sum is greater than zero if and only if at least one term in the sum is greater than zero.
Furthermore, from previous proofs, we know exactly what each term in the sum represents:

- $(I)_{ij} > 0$: Only if $i = j$. This represents a path of 0 steps (staying at the same page)
- $(A)_{ij} > 0$: If there is a path of exactly 1 step from j to i
- $(A^2)_{ij} > 0$: If there is a path of exactly 2 steps from j to i

- $(A^p)_{ij} > 0$: If there is a path of exactly p steps from j to i

Forward Direction : If $(S_p)_{ij} > 0$, then there must be some integer k (where $0 \leq k \leq p$) such that the term $(A^k)_{ij} > 0$. Based on our earlier work, this means there is a path of length k from j to i. Since $k \leq p$, page i can be reached from page j in p or fewer steps.
Backward Direction : If page i can be reached from page j in p or fewer steps, then there exists some path of length k, where $0 \leq k \leq p$. Because there is a path of length k, the entry $(A^k)_{ij}$ must be greater than zero. Since $(A^k)_{ij}$ is one of the components of the sum $(S_p)_{ij}$, and all other components are ≥ 0, the total sum $(S_p)_{ij}$ must be greater than zero.

4.2.4 Explain why

$$I + A + A^2 + \ + A^{n-1}$$

is a positive matrix if the web is strongly connected

A web (or any directed graph) is strongly connected if, for every pair of pages j and i, there exists at least one directed path starting at j and ending at i.In terms of navigation, this means you can get from any page to any other page in the network by following a sequence of at most $n-1$ links. We previously established that:

$$(I + A + A^2 + \cdots + A^{n-1})_{ij} > 0$$

if and only if page i can be reached from page j in $n-1$ or fewer steps.Now, let's look at any arbitrary entry (i, j) in the sum matrix:

- $i = j$ (Diagonal entries): The term $(I)_{ii} = 1$. Since all other terms in the sum are ≥ 0, the sum $(S)_{ii}$ must be at least 1. Thus, all diagonal entries are positive.
- $i \neq j$ (Off-diagonal entries): Because the web is strongly connected, we are guaranteed that a path exists from j to i. As argued in Step 2, this path requires at most $n-1$ steps. This means that for some k where $1 \leq k \leq n-1$, the entry $(A^k)_{ij}$ must be greater than zero.

Since every pair (i, j) has a path of length $k \in \{0, 1, \ldots, n-1\}$, then for every entry in the sum matrix:

$$(S)_{ij} = \underbrace{(I)_{ij} + (A)_{ij} + \cdots + (A^k)_{ij} + \cdots + (A^{n-1})_{ij}}_{\text{At least one of these terms is } >0} > 0$$

Because every entry in the matrix $I + A + A^2 + \cdots + A^{n-1}$ is strictly greater than zero, the matrix is, by definition, positive.

4.2.5 Use the last part (and Exercise 8) so show that

$$B = \frac{1}{n}(I + A + A^2 + \ + A^{n-1})$$

is positive and column-stochastic (and hence by Lemma 3.2, $dim(V_1(B)) = 1$**).**

Non-negativity We have already shown that if the web is strongly connected, then every pair of pages (i, j) is connected by a path of length k, where $0 \leq k \leq n-1$. As established in the previous

step, the sum $S = I + A + A^2 + \cdots + A^{n-1}$ results in a matrix where every entry $S_{ij} > 0$.
Since $B = \frac{1}{n}S$, and n is a positive integer (the number of pages), multiplying by the scalar $\frac{1}{n}$ preserves the sign of the entries.
Therefore, $B_{ij} > 0$ for all i, j, making B a positive matrix
Column sums A matrix is column-stochastic if the sum of the entries in each column equals 1. To show this for B, we need to look at the properties of the adjacency matrix A, which is already columns stochastic. Two key properties of stochastic matrices are:

- Powers: If A is column-stochastic, then any power A^k is also column-stochastic.
- Identity: The identity matrix I is column-stochastic (its columns sum to 1).

Let e be a row vector of ones $[1, 1, \ldots, 1]$. A matrix M is column-stochastic if and only if $eM = e$. Let's check this for B:

$$eB = e\left[\frac{1}{n}(I + A + A^2 + \cdots + A^{n-1})\right]$$

$$eB = \frac{1}{n}(eI + eA + eA^2 + \cdots + eA^{n-1})$$

Since each term $(I, A, A^2, \ldots)$ is column-stochastic, each $eA^k = e$:

$$eB = \frac{1}{n}(\underbrace{e + e + \cdots + e}_{n \text{ times}})$$

$$eB = \frac{1}{n}(n \cdot e) = e$$

Because the sum of each column in B is 1 and all entries are positive, B is a column-stochastic matrix.

4.2.6 Show that if $x \in V_1(A)$ then $x \in V_1(B)$. Why does this imply that $dim(V_1(A)) = 1$?

Show that if $x \in V_1(A)$, then $x \in V_1(B)$
By definition, if $x \in V_1(A)$, then x is an eigenvector of A associated with the eigenvalue $\lambda = 1$. This means:

$$Ax = x$$

If we apply higher powers of A to the vector x, the result remains unchanged:

$$A^2x = A(Ax) = Ax = x$$

$$A^3x = A(A^2x) = Ax = x$$

Generally, $A^kx = x$ for any $k \geq 0$ (where $A^0 = I$).
Now, let's apply the matrix B to the vector x:

$$Bx = \left[\frac{1}{n}(I + A + A^2 + \cdots + A^{n-1})\right]x$$

Distributing x through the sum:

$$Bx = \frac{1}{n}(Ix + Ax + A^2x + \cdots + A^{n-1}x)$$

Substituting $A^k x = x$ for every term:

$$Bx = \frac{1}{n}(\underbrace{x + x + \cdots + x}_{n \text{ terms}})$$

$$Bx = \frac{1}{n}(nx) = x$$

Since $Bx = x$, by definition, $x \in V_1(B)$.
Does it implies that $\dim(V_1(A)) = 1$**?**
We can use the relationship between the two subspaces to bound the dimension of $V_1(A)$.
The proof above shows that every vector in $V_1(A)$ is also in $V_1(B)$. In set notation, this means $V_1(A) \subseteq V_1(B)$. If one subspace is contained within another, its dimension must be less than or equal to the dimension of the larger subspace:

$$\dim(V_1(A)) \leq \dim(V_1(B))$$

In the previous step, we established that B is a positive column-stochastic matrix. According to Lemma 3.2 (Perron-Frobenius), the eigenspace of such a matrix for $\lambda = 1$ is strictly one-dimensional:

$$\dim(V_1(B)) = 1$$

Since $\dim(V_1(A))$ is an integer, and A must have at least one eigenvector for $\lambda = 1$ (because it is stochastic), we are left with:

$$1 \leq \dim(V_1(A)) \leq 1$$

Therefore, $\dim(V_1(A)) = 1$.